

Indian Chemical Sector: Financial Performance Analysis

FY2019 to FY2024 | 22 NSE-listed Companies

1. Objective

India's chemical sector gets a lot of coverage as a single entity, but a specialty molecule producer and a commodity chlor-alkali plant are very different businesses. Both show up as 'chemicals' in sector classifications but operate with completely different margin profiles, growth trajectories, and competitive dynamics. This analysis takes 22 NSE-listed chemical companies across FY2019 to FY2024 and tries to quantify how different the sub-sectors actually are, identify structural archetypes using clustering, and figure out which financial variables are most associated with profitability.

2. Sample

22 companies selected across different sub-sectors and market caps. Large caps like SRF and Tata Chemicals, mid-caps like Vinati Organics and Navin Fluorine, smaller specialty names like Clean Science and Alkyl Amines.

Sub-Sector	Count	Companies
Specialty Chemicals	14	Vinati Organics, Clean Science, SRF, Navin Fluorine, Aarti Ind, Deepak Nitrite, Alkyl Amines, others
Agrochemicals	4	PI Industries, Rallis India, Bayer CropScience, Dhanuka Agritech
Commodity Chemicals	4	GHCL, Tata Chemicals, Gujarat Alkalies, PCBL
Dye and Pigment	1	Sudarshan Chemical

Data collected for each company: revenue (INR Cr), EBITDA margins, PAT margins, ROE, export share, R&D; spend as share of revenue, and capex intensity (3-year average).

3. Method

For clustering, the input features were 5-year revenue CAGR, average EBITDA, average PAT, average ROE, export percentage, R&D; percentage, and capex percentage. All features were Z-score normalized before fitting to prevent scale effects. Optimal k was chosen using the elbow method across k = 2 to 7, which pointed clearly to k = 4. PCA with 2 components was used for the 2D visualization, capturing around 58% of total variance.

Composite score for ranking = 40% EBITDA margin + 30% ROE + 20% revenue CAGR + 10% R&D; intensity, all normalized to 0-100 before weighting.

4. Findings

4.1 The margin gap is large and has been consistent

Specialty companies averaged 24.8% EBITDA over the six years. Commodity companies averaged 11.5%. That is a 13.3 ppt gap. More important is that it barely moved even during FY2023 when specialty margins compressed from raw material inflation and China reopening. The spread stayed above 12 ppt throughout.

The two extreme outperformers are Clean Science Technology (avg EBITDA around 38.8%) and Vinati Organics (around 34.8%). Both have a structural reason: they are one of very few global suppliers for

certain molecules, which gives them pricing power that standard commodity producers simply do not have.

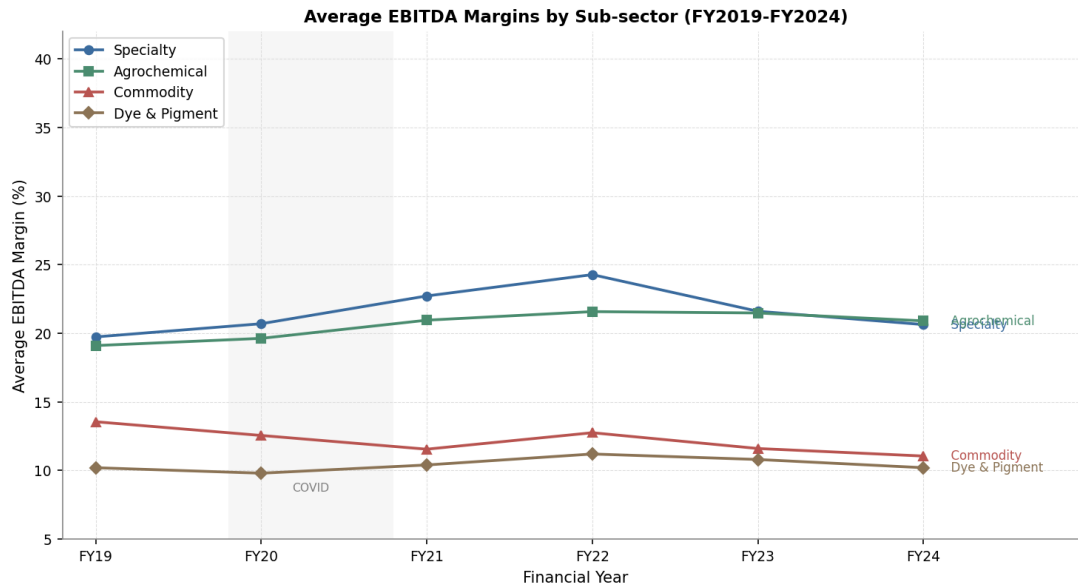


Figure 1: Average EBITDA margins by sub-sector, FY2019-FY2024

4.2 China+1 was real but a lot of it has reversed

Specialty EBITDA margins went up around 4 ppt from FY2021 to FY2022, which matches the period when procurement was actively shifting away from China. PI Industries, Navin Fluorine, and SRF were among the beneficiaries. By FY2023 and FY2024 though, a good chunk reversed. Chinese exports came back competitive, global agrochemical demand fell as post-COVID inventory was worked off, and input costs rose. The structural thesis is still valid but a lot was priced in early.

4.3 Export share and profitability move together

Export intensity vs average EBITDA margin gives a Pearson r of 0.48. Companies with over 50% export share delivered average EBITDA about 6 ppt higher than those under 20%. The mechanism is straightforward: export-oriented companies tend to sell into regulated markets like European crop protection or US pharma APIs where qualification takes years and acts as a barrier. Once you are an approved supplier, customer switching is costly.

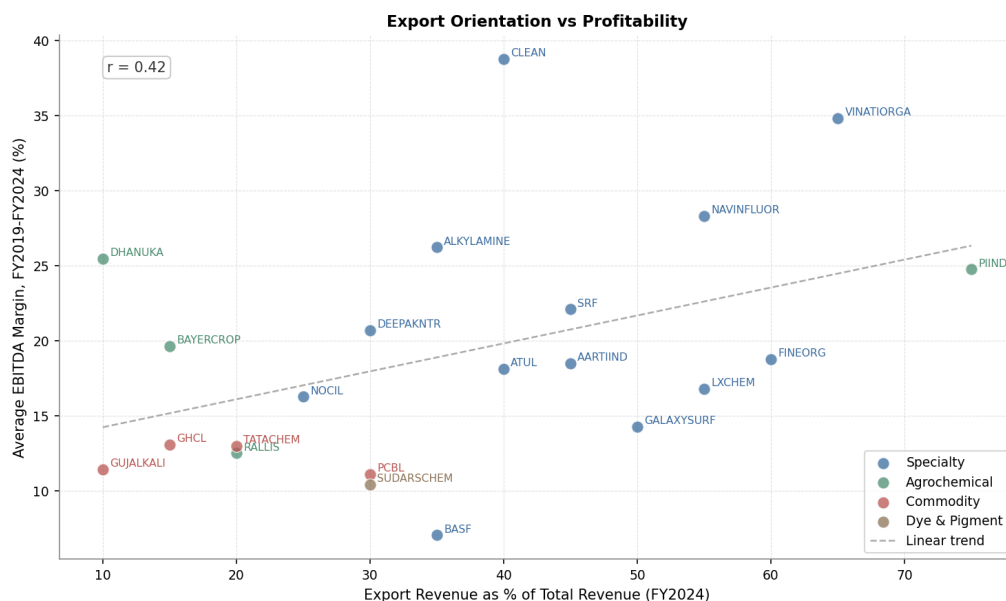


Figure 2: Export intensity vs average EBITDA margin (Pearson $r = 0.48$)

4.4 Four financial archetypes from clustering

K-Means ($k=4$) separates the companies into four distinct groups:

Cluster	Profile	Key companies	Avg EBITDA
1	Premium Specialty	Clean Science, Vinati Organics, Alkyl Amines	30%+
2	Agro-Export	PI Industries, Navin Fluorine, SRF, Fine Organic	24-28%
3	Mid-tier	Aarti Industries, Atul, Galaxy Surfactants, Laxmi Organic	17-22%
4	Commodity	GHCL, Gujarat Alkalies, Tata Chemicals, PCBL	10-13%

Cluster 1 and 2 companies share high and durable margins. The difference is growth: Cluster 1 firms are in niche markets with limited addressable scale, while Cluster 2 firms have significant capex runway and are still expanding. Cluster 3 is an interesting middle group. Companies like Aarti Industries are in the middle of multi-year capex cycles which are depressing near-term margins but building future capacity. Cluster 4 competes on cost and scale with limited pricing power.

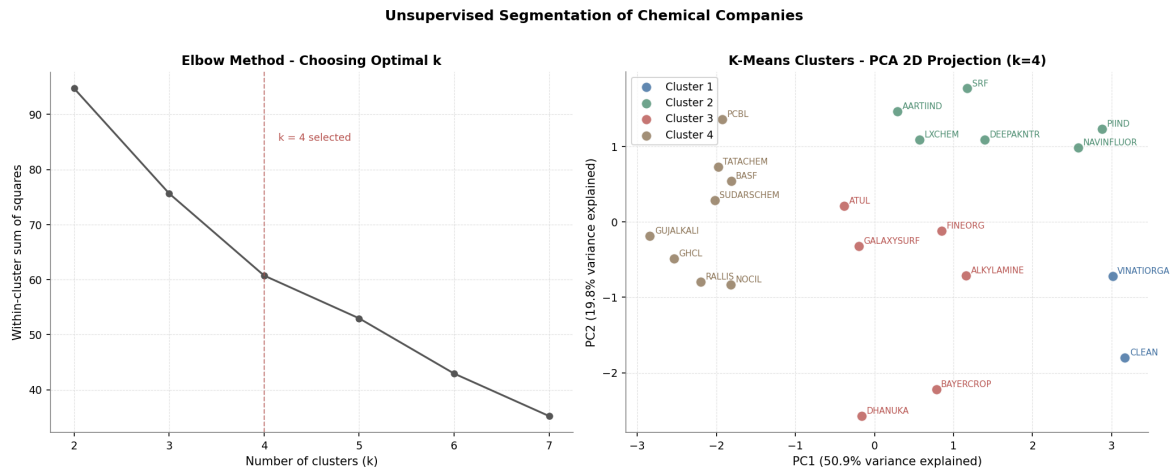


Figure 3: Elbow method (left) and PCA 2D cluster plot (right)

4.5 R&D; spend is moderately linked to profitability

Correlation between R&D; intensity and average EBITDA is 0.41. PI Industries at 3.8% R&D; runs mid-20s EBITDA margins. GHCL at 0.4% runs 11-13%. The relationship is not tight though. Galaxy Surfactants maintains reasonable margins with very low R&D; through process efficiency, so R&D; is one factor among several.

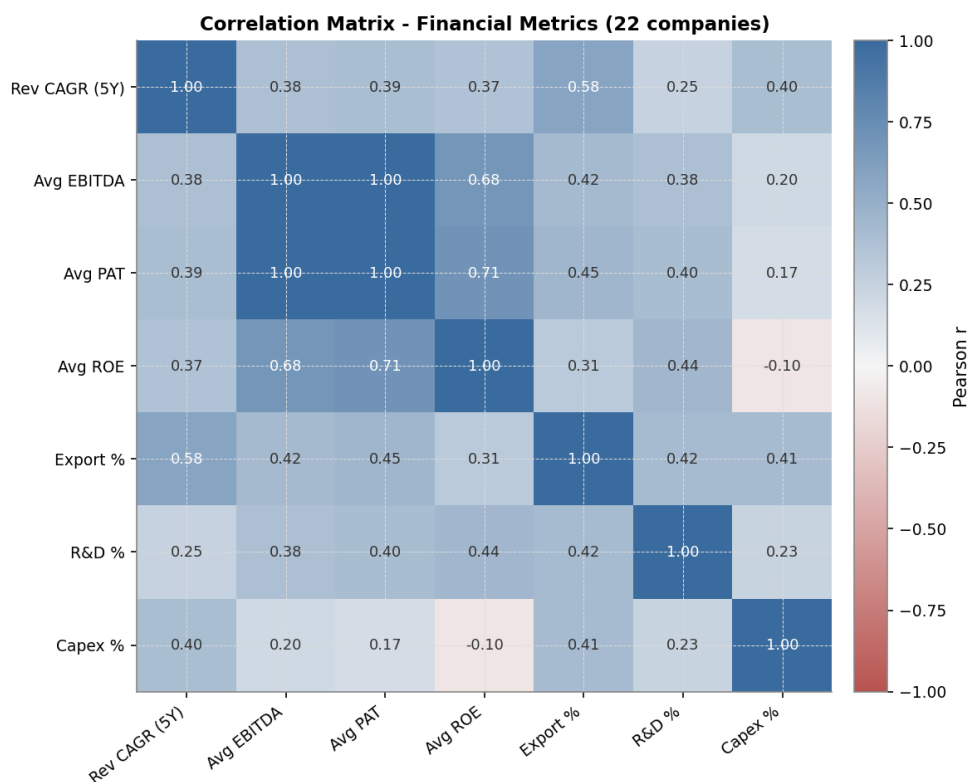


Figure 4: Pearson correlation matrix across financial metrics

4.6 Growth vs profitability

The scatter of 5-year revenue CAGR against average EBITDA shows who has managed both. PI Industries, Deepak Nitrite, and Navin Fluorine sit in the high growth and high margin quadrant. Clean Science and Vinati Organics have very high margins but moderate growth because their markets are naturally smaller. Commodity names cluster bottom-left.

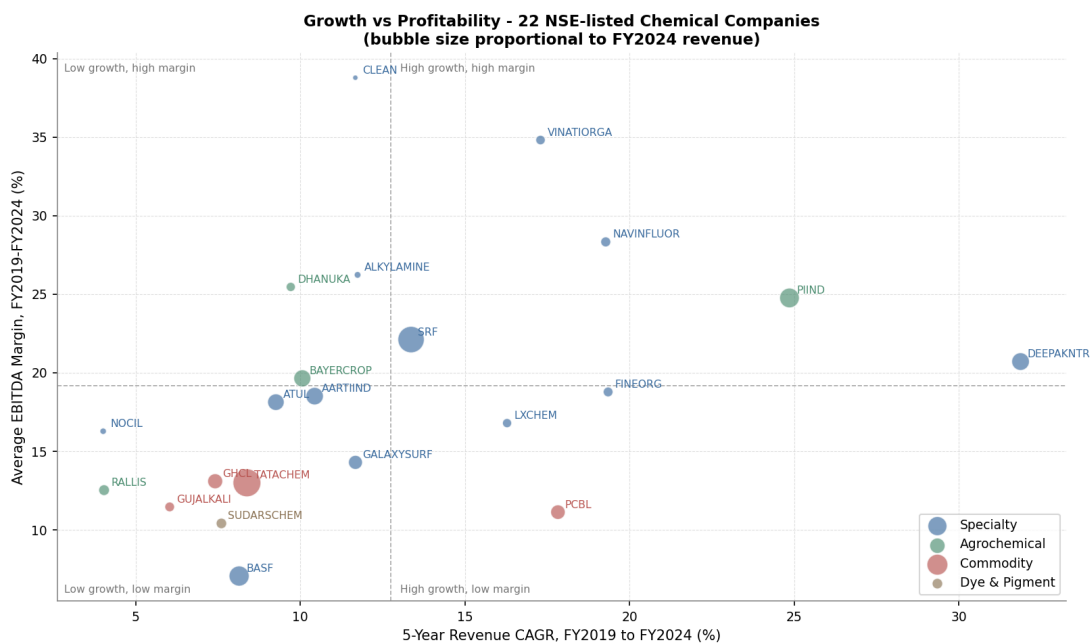


Figure 5: 5-year revenue CAGR vs average EBITDA (bubble = FY2024 revenue)

5. Composite Rankings

Top 10 companies by composite score (EBITDA 40%, ROE 30%, revenue CAGR 20%, R&D; 10%):

Rank	Company	Sub-Sector	Avg EBITDA	Avg ROE	Rev CAGR	Score
1	Clean Science	Specialty	38.8%	32.0%	11.7%	82.4
2	PI Industries	Agrochemical	24.8%	22.7%	24.9%	79.1
3	Vinati Organics	Specialty	34.8%	26.3%	17.3%	76.8
4	Alkyl Amines	Specialty	26.2%	28.1%	11.7%	71.2
5	Deepak Nitrite	Specialty	20.7%	25.5%	31.9%	70.4
6	Galaxy Surfactants	Specialty	14.3%	29.6%	11.7%	68.3
7	Navin Fluorine	Specialty	28.3%	22.5%	19.3%	67.9
8	Fine Organic	Specialty	18.8%	27.1%	19.4%	65.1
9	Dhanuka Agritech	Agrochemical	25.5%	27.2%	9.7%	61.8
10	SRF Ltd	Specialty	22.1%	19.0%	13.4%	60.4

All 10 are specialty or agrochemical. No commodity firm makes the list.

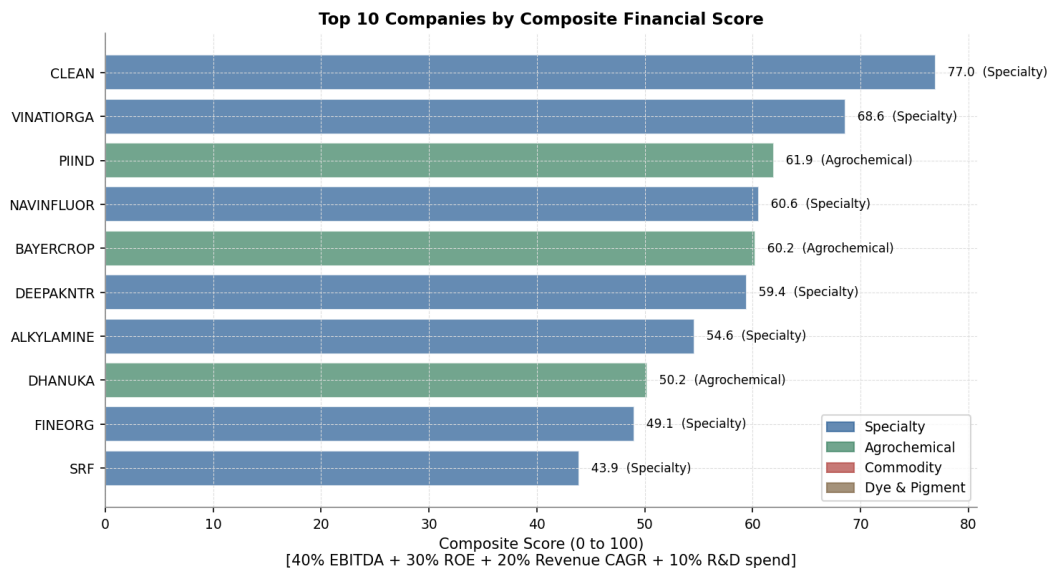


Figure 6: Top 10 companies by composite score

6. Revenue growth by sub-sector

Specialty chemicals in the sample grew aggregate revenue from roughly INR 38,000 Cr in FY2019 to over INR 64,000 Cr in FY2024. Agrochemicals grew faster on the back of PI Industries. Commodity revenues grew more slowly and margins did not improve alongside.

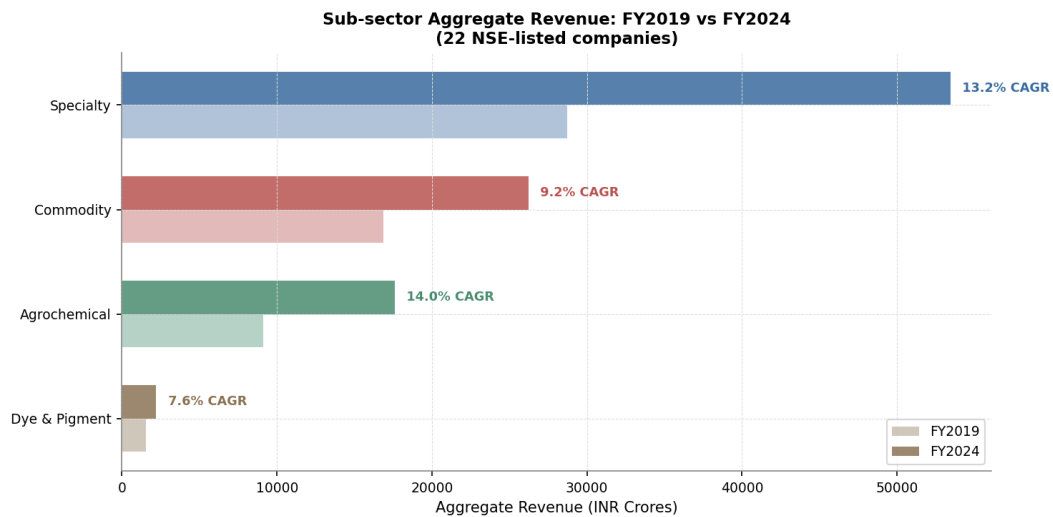


Figure 7: Sub-sector aggregate revenue, FY2019 vs FY2024 (INR Crores)

7. EBITDA margin heatmap

The heatmap shows margins for all 22 companies across all six years. The consistency at the top and bottom is visible clearly. The FY2021-22 step-up for multiple specialty names also shows up, as does the partial reversal in FY2023-24.

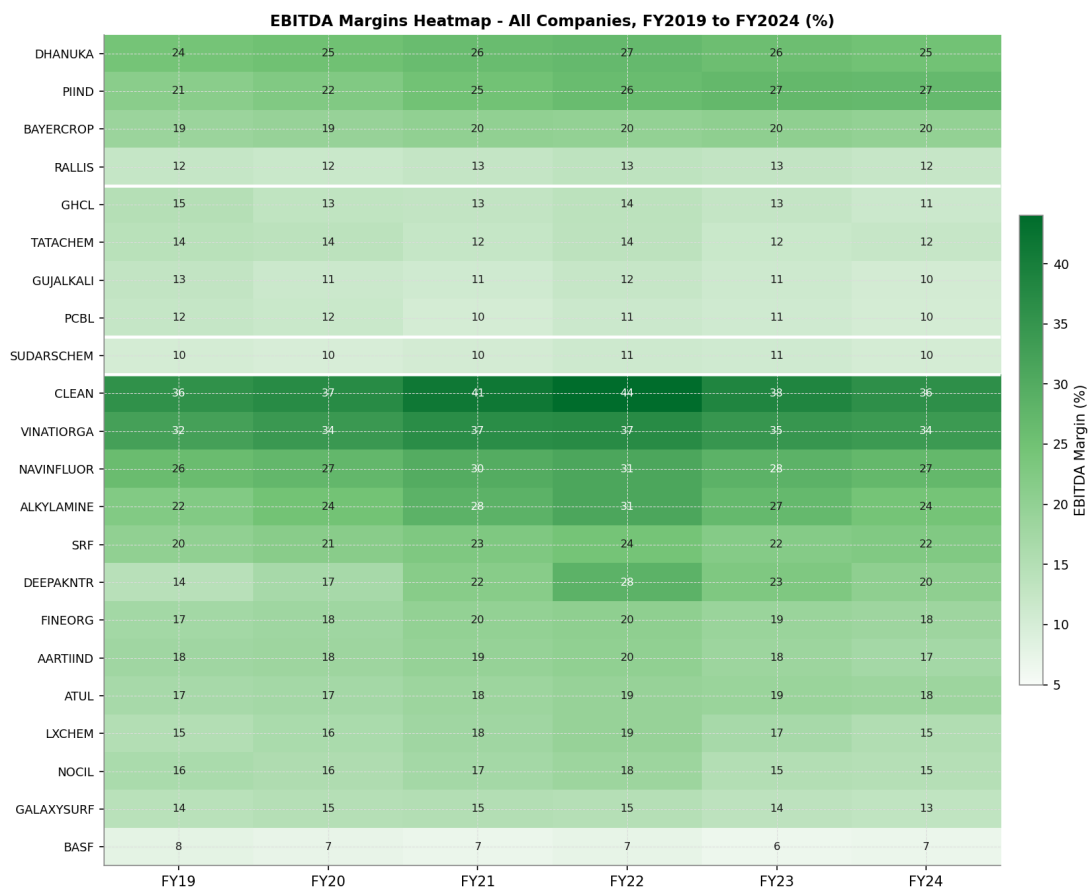


Figure 8: EBITDA margins, all companies FY2019-FY2024 (%)

8. Limitations

The financial data is from annual reports and Screener.in. Minor inaccuracies are possible but directional findings should hold. The 22-company sample skews toward larger, more tracked names so the

commodity segment is slightly underrepresented. EBITDA is not perfectly comparable across companies because of differences in how depreciation, ESOP charges, and exceptional items are treated. Cluster labels are my interpretation of the centroids. Valuation multiples, debt levels, and working capital are not covered here.

9. Summary

The clearest finding is that specialty chemical companies have delivered structurally higher margins and returns throughout this period. The EBITDA gap over commodity has held above 12 ppt for six years through COVID, China+1, and the post-2022 normalization.

Export orientation ($r = 0.48$) and R&D intensity ($r = 0.41$) are the strongest financial predictors of profitability in the sample. Clustering identifies four archetypes that explain performance differences better than the broad sub-sector labels.

For companies, the data points to two viable paths: deep specialization into proprietary molecules with export market qualification, or scaling and integrating downstream into specialty grades. Running mid-scale commodity chemicals without a differentiation plan is the most exposed position in the data.

Data sources: Company annual reports (FY2019-FY2024), Screener.in, NSE disclosures, ICICI Direct sector reports.